# Small signal staility analysis on large systems

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# Introduction

An EMT system defined in the *abc* domain is characterized by periodic sinusoidal voltages and currents. These periodic signals do not allow small-signal stability analysis calculations, since this analysis can only be done with constant steady state values for all the signals in the system. However, it is known that this sinusoidal perturbation in the *abc* domain is the correct steady state behavior of an AC network [1, 2].

Other stability analysis methods are needed to consider these natural perturbations in the EMT simulations in the *abc* domain. The Floquet theory is considered and exposed in the following sections.

# 1 Stability analysis on periodic linear systems

## 1.1 Floquet Theory

Consider a linear time-varying system of the form

$$\dot{x}(t) = A(t)\, x(t), \tag{1}$$

where $x(t) \in R^n$ and the system matrix $A(t)$ is *periodic* with period $T$, i.e. [3, 4] .

$$A(t+T) = A(t), \qquad \forall t. \tag{2}$$

Such systems naturally appear in many engineering applications, including electrical power systems, rotating machines, and systems described in a rotating reference frame. Floquet theory provides a systematic framework to analyse the behavior and stability of these systems.

### 1.1.1 Fundamental Matrix Solution

Let $M(t)$ denote the *fundamental matrix* of the system, defined as the matrix solution of

$$\dot{M}(t) = A(t)\, M(t), \qquad M(0) = I, \tag{3}$$

where $I$ is the identity matrix. The fundamental matrix describes how infinitesimal perturbations evolve over time, since the solution of the system with initial condition $x(0)$ can be written as

$$x(t) = M(t)\, x(0). \tag{4}$$

### 1.1.2 Floquet Theorem

Floquet theory states that the fundamental matrix of a linear periodic system can always be written as

$$M(t) = P(t)\, e^{Rt}, \tag{5}$$

where:

- $P(t)$ is a periodic matrix with period $T$, i.e. $P(t+T) = P(t)$,
- $R$ is a constant matrix [5] .

This decomposition shows that the system dynamics can be separated into a purely exponential part, governed by $R$, and a periodic modulation, represented by $P(t)$. Importantly, all long-term stability properties of the system are determined by the exponential term.

### 1.1.3 Monodromy Matrix

Evaluating the fundamental matrix after one full period yields

$$C_M := M(T), \tag{6}$$

which is known as the *monodromy matrix* [5]. Using the Floquet decomposition,

$$C_M = M(T) = P(T)\, e^{RT} = P(0)\, e^{RT}, \tag{7}$$

since $P(T) = P(0)$. Therefore, the eigenvalues of $C_M$ are given by

$$\mu_i = e^{\lambda_i T}, \tag{8}$$

where $\lambda_i$ are the eigenvalues of $R$. The quantities $\mu_i$, the eigenvalues of $C_M$, are called the *Floquet multipliers* which are the ones used for the stability analysis.

### 1.1.4 Stability Criterion

The stability of the periodic system is completely determined by the Floquet multipliers [6] :

- If $|\mu_i| < 1$ for all $i$, the system is asymptotically stable.
- If $|\mu_i| = 1$ for at least one $i$ and all others satisfy $|\mu_j| \leq 1$, the system is marginally stable.
- If $|\mu_i| > 1$ for at least one $i$, the system is unstable.

Thus, stability analysis of a linear periodic system reduces to computing the eigenvalues of the monodromy matrix.

### 1.1.5 Practical Remarks

The fundamental matrix $M(t)$ must be obtained by integrating the matrix differential equation $\dot{M}(t) = A(t)M(t)$.

If the matrices commute ($A(t_1)A(t_2) = A(t_2)A(t_1)$),the fundamental matrix can be computed as

$$M(t) = \exp\left(\int_0^t A(\tau)\, d\tau\right)$$

This numerical integration step is essential in practical applications of Floquet theory.

## 1.2 From DAE to Floquet stability analysis

The DAEs of the system expressed explicitly as

$$\begin{aligned} \dot{x} &= f(x, y, t) \\ 0 &= g(x, y, t) \end{aligned} \tag{9}$$

Is refactored into the linearized state-space representation

$$\dot{x}(t) = A(t)x(t) \tag{10}$$

with the state matrix $A(t)$ being

$$A(t) = \frac{\partial f(t)}{\partial x} - \left(\frac{\partial f(t)}{\partial y}\ \left(\frac{\partial g(t)}{\partial y}\right)^{-1}\ \frac{\partial g(t)}{\partial x}\right) \tag{11}$$

Afterwards the fundamental matrix ODE ($\dot{M}(t) = A(t)\ M(t)$) is integrated from 0 to T, being T the period of the oscillations and evaluated at t=T to find the monodromy matrix $C_M$

$$C_M = M(T) = \int_0^T \dot{M}(\tau)d\tau = \int_0^T A(\tau)\ M(\tau)d\tau \tag{12}$$

Finally eigenvalues from the monodromy matrix are computed and the stability analysis is performed.

## 1.3 Modal Interpretation: Damping Ratios and Participation Factors

While the Floquet multipliers are sufficient to assess the stability of a linear periodic system, additional insight can be obtained by analysing modal properties analogous to those used in classical small-signal stability analysis, namely damping ratios and participation factors. Although these concepts are not defined in exactly the same way as in time-invariant systems, meaningful generalisations exist within the Floquet framework [7]
.

### 1.3.1 Floquet Exponents and Damping Ratios

The eigenvalues of the monodromy matrix $C_M$, denoted as $\mu_i$, are related to the so-called *Floquet exponents* $\lambda_i$ through

$$\lambda_i = \frac{1}{T}\ln(\mu_i), \tag{13}$$

where $T$ is the period of the system and the complex logarithm is evaluated with an appropriate branch selection. Each Floquet exponent can be written as

$$\lambda_i = \alpha_i + j\beta_i, \tag{14}$$

where $\alpha_i$ represents the average exponential growth or decay rate per unit time, and $\beta_i$ corresponds to an average oscillatory frequency.

In direct analogy with small-signal analysis of linear time-invariant systems, a *Floquet damping ratio* can be defined as

$$\zeta_i^{\mathrm{F}} = -\frac{\alpha_i}{\sqrt{\alpha_i^2 + \beta_i^2}}. \tag{15}$$

This damping ratio represents an average measure of modal damping over one period of the oscillation. It must be emphasised that, unlike the LTI case, this quantity does not describe instantaneous damping, but rather the net effect of the periodic dynamics over a full cycle.

### 1.3.2 Participation Factors in Floquet Systems

In linear time-invariant systems, participation factors quantify the contribution of each state variable to a given mode using the left and right eigenvectors of the system matrix. In periodic systems, the system matrix $A(t)$ and its eigenvectors are time-dependent, which prevents a unique instantaneous definition of participation factors.

A practical and widely used approach is to define participation factors based on the eigenstructure of the monodromy matrix. Let $r_i$ and $l_i$ denote the right and left eigenvectors of $C_M$ associated with the Floquet multiplier $\mu_i$, satisfying

$$C_M r_i = \mu_i r_i, \qquad l_i^\top C_M = \mu_i l_i^\top. \tag{16}$$

The participation factor of state $k$ in mode $i$ is then defined as

$$p_{ki}^{\mathrm{F}} = r_{k,i}\, l_{i,k}, \tag{17}$$

where $r_{k,i}$ is the $k$-th component of the right eigenvector and $l_{i,k}$ is the $k$-th component of the left eigenvector.

These participation factors describe how strongly each state contributes to the growth or decay of perturbations over one period. Although they are evaluated at a specific reference time (typically $t = 0$), they provide valuable modal insight comparable to classical participation factors.

### 1.3.3 Interpretation and Limitations

The damping ratios and participation factors obtained from Floquet theory should be interpreted as *period-averaged modal properties*. They are particularly useful for identifying poorly damped periodic modes and for understanding which states are most influential in the stability of EMT simulations expressed in the *abc* domain.

However, it must be noted that, due to the inherent time dependence of periodic systems, these quantities do not capture instantaneous modal behaviour. Their interpretation is therefore global in time, rather than local, and should be used accordingly.

## 1.4 Scalable Floquet Stability Analysis for Large-Scale Systems

The direct computation of the monodromy matrix $C_M \in R^{n\times n}$ requires storing and manipulating dense matrices, leading to memory and computational costs of order $\mathcal{O}(n^2)$ and $\mathcal{O}(n^3)$, respectively. For large-scale EMT models, where the number of state variables can reach several thousands or more, this approach becomes impractical. As a consequence, large-scale Floquet analysis relies on matrix-free techniques that avoid the explicit construction of $C_M$ [4, 5].

### 1.4.1 Directional Variational Equations

Consider again the linear periodic system

$$\dot{x}(t) = A(t)\, x(t), \tag{18}$$

and its associated fundamental matrix equation

$$\dot{M}(t) = A(t)\, M(t), \qquad M(0) = I. \tag{19}$$

Each column of the fundamental matrix $M(t)$ evolves independently according to the same linear system. This observation leads to the definition of the *directional variational equation*. Given an initial perturbation direction $v_0 \in R^n$, the directional variational equation is defined as

$$\dot{v}(t) = A(t)\, v(t), \qquad v(0) = v_0. \tag{20}$$

After integrating this equation over one period, the resulting vector satisfies

$$v(T) = C_M\, v_0, \tag{21}$$

which corresponds to the action of the monodromy matrix on the initial direction. Therefore, by integrating the variational equation for different initial directions, the effect of the monodromy matrix can be evaluated without explicitly forming it [4].

### 1.4.2 Monodromy Operator Formulation

Instead of viewing the monodromy matrix as an explicit matrix, it is convenient to interpret it as a linear operator

$$\mathcal{C} : R^n \to R^n, \qquad \mathcal{C}(v_0) = v(T), \tag{22}$$

where $v(T)$ is obtained by integrating the directional variational equation over one period. This operator-based formulation enables a matrix-free representation of the monodromy map, which is particularly suitable for large-scale systems and is standard in numerical Floquet analysis [5].

### 1.4.3 Krylov Subspaces

To compute only the most critical Floquet multipliers, it is not necessary to determine the full spectrum of $\mathcal{C}$. Instead, iterative eigensolvers based on Krylov subspaces can be employed. Given an initial vector $v_0$, the Krylov subspace of dimension $m$ associated with the monodromy operator is defined as

$$\mathcal{K}_m(\mathcal{C}, v_0) = \operatorname{span}\left\{v_0,\ \mathcal{C}(v_0),\ \mathcal{C}^2(v_0),\ \ldots,\ \mathcal{C}^{m-1}(v_0)\right\}. \tag{23}$$

Krylov subspace methods are widely used in large-scale eigenvalue problems, as they allow accurate approximation of the dominant eigenvalues of a linear operator using a subspace of dimension $m \ll n$ [8].

### 1.4.4 Arnoldi Method for Floquet Analysis

The Arnoldi method is a standard Krylov-based algorithm used to construct an orthonormal basis of the Krylov subspace while projecting the operator $\mathcal{C}$ onto a low-dimensional Hessenberg matrix [8]. Applied to Floquet analysis, the algorithm proceeds as follows:

- An initial perturbation direction $v_0$ is selected and normalised.
- The directional variational equation is integrated over one period to compute $\mathcal{C}(v_0)$.
- The resulting vector is orthogonalised with respect to the previously computed basis vectors.
- This procedure is repeated to build an orthonormal basis of the Krylov subspace.

At each iteration, the action of the monodromy operator is obtained solely through time-domain integration of the variational equation. The Arnoldi process yields a small projected matrix whose eigenvalues approximate the dominant Floquet multipliers of the original system [5, 9].

### 1.4.5 Advantages for Large-Scale EMT Systems

The combination of directional variational equations and Arnoldi-based Krylov methods provides a scalable framework for Floquet stability analysis with the following advantages:

- No explicit construction or storage of the monodromy matrix is required.
- Memory requirements scale as $\mathcal{O}(nm)$ instead of $\mathcal{O}(n^2)$.
- Only a limited number of dominant Floquet multipliers need to be computed.
- The approach integrates naturally with existing time-domain EMT solvers.

This methodology has been successfully applied to large-scale periodic power system models and EMT simulations, where classical Floquet techniques based on full monodromy matrices are computationally infeasible [7, 9].

## 1.5 Reduced-Order Participation Analysis in Large-Scale Floquet Systems

When matrix-free Krylov methods are employed to compute the dominant Floquet multipliers of large-scale periodic systems, the full monodromy matrix is not explicitly constructed. As a consequence, classical participation factors, which require access to the complete set of left and right eigenvectors of the monodromy matrix, cannot be directly computed for the full state space.

To overcome this limitation while retaining scalability, a reduced-order participation analysis can be performed by focusing on a subset of state variables that are most relevant to the dominant Floquet modes. This approach enables the computation of classical participation factors on a reduced subsystem, while avoiding the prohibitive computational cost of full-scale Floquet analysis.

### 1.5.1 Selection of a Reduced Subsystem

The first step consists of identifying a reduced set of state variables that significantly contribute to the dominant Floquet modes. This selection can be guided by:

- the magnitude of the components of the approximate Floquet eigenvectors obtained from the Krylov subspace,
- energy-based participation measures computed from directional variational trajectories,
- or physical insight into the system dynamics (e.g., converter control states, generator internal variables).

Let $\mathcal{S} \subset \{1, \ldots, n\}$ denote the index set of the selected state variables, with cardinality $n_r \ll n$. The reduced state vector is then defined as

$$x_r(t) = \Pi x(t), \tag{24}$$

where $\Pi \in R^{n_r \times n}$ is a projection matrix that extracts the selected states.

### 1.5.2 Construction of the Reduced Periodic System

Restricting the original periodic system

$$\dot{x}(t) = A(t)x(t) \tag{25}$$

to the reduced state vector yields the reduced-order system

$$\dot{x}_r(t) = A_r(t)x_r(t), \tag{26}$$

where the reduced periodic state matrix is given by

$$A_r(t) = \Pi A(t)\Pi^\top. \tag{27}$$

This reduced system retains the dominant dynamic interactions associated with the selected Floquet modes, while significantly reducing the system dimension.

### 1.5.3 Local Monodromy Matrix

For the reduced-order system, the fundamental matrix $M_r(t)$ is defined as

$$\dot{M}_r(t) = A_r(t)M_r(t), \qquad M_r(0) = I_{n_r}, \tag{28}$$

where $I_{n_r}$ is the identity matrix of dimension $n_r$.

Integrating this equation over one period yields the *local monodromy matrix*

$$C_{M,r} = M_r(T), \tag{29}$$

which approximates the action of the full monodromy matrix on the dominant invariant subspace.

### 1.5.4 Computation of Participation Factors

Since the reduced monodromy matrix $C_{M,r} \in R^{n_r \times n_r}$ is explicitly available, classical modal analysis can be performed. Let $\mu_i$ denote a dominant Floquet multiplier of the reduced system, with associated right and left eigenvectors $r_i$ and $l_i$, satisfying

$$C_{M,r} r_i = \mu_i r_i, \qquad l_i^\top C_{M,r} = \mu_i l_i^\top. \tag{30}$$

The participation factor of reduced state $k$ in Floquet mode $i$ is then defined as

$$p_{ki}^{(r)} = r_{k,i}\, l_{i,k}. \tag{31}$$

These participation factors provide a physically meaningful measure of the contribution of each selected state to the dominant periodic modes, while remaining computationally tractable for large-scale EMT systems.

### 1.5.5 Interpretation and Scope

The reduced-order participation factors should be interpreted as *local* modal measures associated with the dominant Floquet dynamics. While they do not capture the complete modal structure of the full system, they offer valuable insight into the origin and nature of poorly damped periodic modes, enabling targeted control design and model refinement.

# 2 Reduced Participation Factor Analysis in Large-Scale Small-Signal Models

In classical small-signal stability analysis, participation factors are widely used to identify the contribution of individual state variables to specific dynamic modes [7, 10]. For large-scale power system models, however, the computation of participation factors based on the full state matrix may be computationally demanding or unnecessary. As a result, reduced-order participation factor analysis is commonly adopted in practice [1].

## 2.1 Small-Signal Linearisation

Consider a nonlinear dynamic system described by

$$\dot{x} = f(x), \tag{32}$$

operating at an equilibrium point $x^\star$. Linearisation around this operating point yields the linear time-invariant system

$$\dot{\Delta x}(t) = A\,\Delta x(t), \tag{33}$$

where

$$A = \left.\frac{\partial f}{\partial x}\right|_{x=x^\star} \tag{34}$$

is the state matrix. The eigenvalues $\lambda_i$ of $A$ determine the small-signal stability of the system, with stability requiring $\mathrm{Re}(\lambda_i) < 0$ for all modes [1].

## 2.2 Classical Participation Factors

Let $r_i$ and $l_i$ denote the right and left eigenvectors of $A$ associated with eigenvalue $\lambda_i$, satisfying

$$Ar_i = \lambda_i r_i, \qquad l_i^\top A = \lambda_i l_i^\top, \tag{35}$$

with the normalisation condition $l_i^\top r_j = \delta_{ij}$. The participation factor of state $k$ in mode $i$ is defined as [10]

$$p_{ki} = r_{k,i}\, l_{i,k}. \tag{36}$$

Participation factors provide a quantitative measure of the influence of each state variable on a given dynamic mode and are extensively used to interpret oscillatory behaviour in power system models [7].

## 2.3 Reduced-Order Participation Analysis

In large-scale systems, it is often neither necessary nor practical to compute participation factors for all state variables. Instead, a reduced-order analysis is performed by restricting the system to a subset of relevant states that are expected to play a dominant role in the dynamics of interest [1].

Let $\mathcal{S} \subset \{1, \ldots, n\}$ denote the index set of selected state variables, with $n_r = |\mathcal{S}| \ll n$. Defining the reduced state vector

$$x_r = \Pi x, \tag{37}$$

where $\Pi \in R^{n_r \times n}$ is a projection matrix, the reduced linear system becomes

$$\dot{x}_r = A_r x_r, \qquad A_r = \Pi A \Pi^\top. \tag{38}$$

Classical eigenvalue and eigenvector analysis can then be applied to the reduced state matrix $A_r$, allowing the computation of reduced participation factors

$$p_{ki}^{(r)} = r_{k,i}^{(r)}\, l_{i,k}^{(r)}, \tag{39}$$

where $r_i^{(r)}$ and $l_i^{(r)}$ denote the right and left eigenvectors of $A_r$, respectively.

### 2.3.1 Scope and Interpretation

Reduced participation factors provide physically meaningful insight into the dominant contributors to specific modes, while maintaining computational tractability for large-scale systems. Although they do not represent the complete modal structure of the full model, they are widely used in practical power system studies to support stability assessment, controller tuning, and model interpretation [1, 7].

## 2.4 Reduced Participation Factors via Krylov Projection

In large-scale power system models, the direct computation of all eigenvalues, eigenvectors, and participation factors from the full state matrix $A \in R^{n\times n}$ becomes computationally prohibitive, since dense eigendecomposition scales as $\mathcal{O}(n^3)$ [11]. For this reason, reduced-order techniques based on Krylov subspaces are commonly employed to extract only the dominant dynamic modes [8].

This section describes the computation of *reduced participation factors* using a projection-based approach and explains why this method remains sparse and scalable, despite involving dense operations in a low-dimensional subspace.

### 2.4.1 Krylov Subspace Projection

Consider the linearised state-space model

$$\dot{x}(t) = Ax(t), \tag{40}$$

where $A$ is sparse and $n$ is large. Using an Arnoldi-based Krylov method, the $k \ll n$ dominant right eigenpairs of $A$ are approximated by solving

$$AV \approx V\Lambda, \tag{41}$$

where $V \in C^{n\times k}$ spans a Krylov subspace constructed using sparse matrix–vector products [8].

The computational complexity of this step scales as

$$\mathcal{O}(k \cdot \mathrm{nnz}(A)),$$

where $\mathrm{nnz}(A)$ denotes the number of nonzero entries of $A$.

### 2.4.2 Reduced Projected System

Once the Krylov basis $V$ is obtained, the system matrix is projected onto the reduced subspace using a Rayleigh–Ritz projection:

$$A_r = V^H AV, \tag{42}$$

where $A_r \in C^{k\times k}$ is the reduced-order system matrix [8].

Although $A_r$ is dense, its dimension depends only on $k$, which is independent of the original system size $n$.

A dense eigendecomposition of the reduced system is then performed:

$$A_r V_r = V_r \Lambda_r, \qquad W_r^H A_r = \Lambda_r W_r^H. \tag{43}$$

### 2.4.3 Lifting of Left Eigenvectors

The left eigenvectors of the original system are approximated by lifting the reduced left eigenvectors back to the full state space:

$$W = VW_r. \tag{44}$$

This construction guarantees the bi-orthogonality condition

$$W^H V = I_k, \tag{45}$$

which is required for the correct definition of participation factors [12].

### 2.4.4 Reduced Participation Factors

The participation factor of state $j$ in mode $i$ is defined as

$$p_{j,i} = |v_{j,i}\, w_{j,i}|\,, \tag{46}$$

which quantifies the relative contribution of each state to a given mode [12].

The participation factors are normalised per mode as

$$\tilde{p}_{j,i} = \frac{p_{j,i}}{\sum_{j=1}^{n} p_{j,i}}, \tag{47}$$

yielding a reduced participation factor matrix $\tilde{P} \in R^{n\times k}$.

### 2.4.5 Scalability Considerations

Although dense matrices appear in the reduced-order computations, the proposed method remains sparse and scalable because all operations whose computational cost depends on the original system dimension $n$ rely exclusively on sparse matrix-vector products. Dense operations are strictly confined to a low-dimensional subspace whose size depends only on the number of retained modes $k$, which is chosen independently of $n$.

The dominant computational step is the construction of the Krylov subspace using an Arnoldi-based method, which requires repeated evaluations of the form

$$y = Ax, \tag{48}$$

where $A \in R^{n\times n}$ is sparse. The cost of each such operation scales as $\mathcal{O}(\mathrm{nnz}(A))$, and the overall cost of the Arnoldi process scales as

$$\mathcal{O}(k \cdot \mathrm{nnz}(A)), \tag{49}$$

where $\mathrm{nnz}(A)$ denotes the number of nonzero entries of $A$ [8]. Importantly, no dense matrix factorizations of size $n \times n$ are ever performed.

Once the Krylov basis $V \in C^{n\times k}$ is obtained, the system matrix is projected onto the reduced subspace, yielding the reduced matrix

$$A_r = V^H A V \in C^{k\times k}. \tag{50}$$

Although $A_r$ is dense, its dimension depends only on $k$, which is typically small (e.g. $k =$ 10–50) and remains constant as the system size $n$ increases. The dense eigendecomposition of $A_r$ therefore has computational complexity

$$\mathcal{O}(k^3), \tag{51}$$

which is negligible compared to the sparse Krylov step for large-scale systems.

The lifting of the reduced left eigenvectors back to the full space,

$$W = VW_r, \tag{52}$$

and the subsequent computation of participation factors involve matrix operations whose cost scales as $\mathcal{O}(nk)$, which is linear in the number of states and therefore scalable for large $n$.

As a result, all computationally expensive operations scale either linearly with the number of nonzero elements of $A$ or depend only on the reduced dimension $k$. The method avoids any operation whose complexity grows as $\mathcal{O}(n^2)$ or $\mathcal{O}(n^3)$, which would otherwise make the analysis infeasible for large EMT or RMS power system models.

For this reason, the projection-based computation of reduced participation factors is considered a sparse and scalable approach, despite the presence of dense computations in the reduced subspace. This interpretation of sparsity is standard in numerical linear algebra and large-scale stability analysis [8, 11].

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